

# Vision-language model-based intelligent assistant for onsite construction safety inspection

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## ABSTRACT

Construction site inspection demands a contextual understanding of dynamic job-site conditions, traditionally relying on inspectors' expertise combined with adherence to predefined safety regulations and industry standards to identify hazards. While vision-language models can detect and describe hazards, they struggle to correlate observations with regulations due to limitations in geometric reasoning. Recent studies show progress in compliance checking, but these models are still challenged by dynamic scenarios. This paper introduces a hybrid framework that combines geometric reasoning with LLM-powered interpretation to improve regulation-aware hazard detection. The system achieved high detection accuracy: 97 % for ladder use, 94.6 % for mobile scaffolding, and 99 % for fire-related work. Captioning performance evaluated through BLEU, ROUGE, METEOR, and BERT Score showed strong semantic alignment. User feedback confirmed its efficiency and ease of use, even under dynamic conditions. By integrating visual data with regulatory reasoning, the proposed system offers a practical, domain-adapted solution for enhancing construction safety inspections.

## 1. Introduction

The construction industry is consistently recognized as one of the most hazardous sectors, characterized by persistently high workplace injury and fatality incidence rates. Despite the implementation of safety training methods, adjustments to safety regulations, improvements in inspection procedures, and extensive research efforts, the prevalence of accidents remains alarmingly high [1,2]. The Bureau of Labor Statistics reports a work-related fatality rate of 3.7 per 100,000 full-time equivalent workers in the United States for 2023, reflecting a significant 5.7 % increase from the previous year [3]. These statistics, beyond the tragic human cost, represent a significant financial burden for individuals, companies, and society at large.

Construction safety inspectors play a critical role in ensuring safe work environments. They usually walk around the jobsite to identify and mitigate hazards before work begins and ensure compliance with safety rules during operations. These safety inspection processes rely heavily on predefined regulations and the prior knowledge of inspectors [4]. This conventional way of inspection restricts the ability to detect unforeseen dangers and address intricate safety issues, particularly in the

context of modern construction sites with their inherent dynamic and complex nature [5]. Furthermore, the growing size and complexity of these sites further compound the challenges faced by inspectors [6]. Even experienced professionals may overlook critical aspects due to these inherent complexities of construction jobsites. The limitations of manual observation are becoming increasingly apparent, highlighting the need for innovative solutions that provide an easy-to-adopt solution for inspectors, which not only inspect the situation but also provide a platform to discuss the hazard mitigation solutions.

Emerging technologies hold immense potential to transform the safety monitoring process by automating hazard detection and information dissemination. Particularly, computer vision has recently attracted increasing attention because of its potential to overcome the limitations of manual observation in identifying on-site hazards. By extracting visual information from on-site images and videos through the integration of computer vision and deep learning techniques, computer vision systems enable continuous and automated detection of unsafe worker behaviors and hazardous working situations [7]. Despite advancements in vision solutions, a critical gap remains in understanding and reasoning based on relevant safety regulations within

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captured visuals. While visual information data offers numerous benefits for construction safety, its practical implementation remains limited [8]. This limitation hinders their full potential in addressing complex safety challenges. Current Computer Vision (CV) systems can classify construction situations as safe or unsafe according to regulations, however they fall short in providing further interpretation and detailed guidance such as the specific mitigation strategies of these hazards like those offered by experienced safety inspectors.

One potential solution lies in the use of Large Language Models (LLMs). LLMs are a specific type of artificial neural network (ANN) trained on massive datasets of text by leveraging techniques like deep learning and transformer architecture to process and understand the nuances of human language [9]. Deep learning employs a layered ANN structure inspired by the human brain, allowing LLMs to extract complex patterns and relationships from data. Transformer architecture enables them to analyze the relationships between words, giving them a strong grasp of context and meaning. LLMs excel at tasks like answering open-ended questions by analyzing the context and information available to them, providing insightful and relevant responses. These features make LLMs well-suited to interpret visual information for better understanding of complex jobsite situations in the construction industry. Current vision-language models (VLMs) (e.g., GPT-4 V, BLIP-2) can extract information from images and describe visual scenes, but they lack the geometric reasoning required for critical safety measurements, making them inadequate for regulatory compliance in complex construction environments [10]. They also fall short in domain-specific reasoning, limiting their ability to translate raw visual data into actionable safety insights. Although custom models can detect and describe hazards, they frequently fail to perform the spatial analysis required for evaluating compliance with safety regulations. While CV techniques are effective in detecting objects and classifying scenes, they lack the capacity to perform higher-level regulatory reasoning. Conversely, VLMs excel at contextual language understanding but cannot execute geometric measurements from visual data. This complementary gap underscores the need for a hybrid framework that integrates both approaches for comprehensive safety inspection.

Building on the identified research gaps, this study is guided by the following central question: How can the integration of geometric reasoning from computer vision and large language models enhance scene understanding for real-time hazard detection and safety compliance on construction sites? To address this challenge, the paper introduces a system built on a novel hybrid reasoning architecture. The primary contribution lies not in developing a new vision model or LLM, but in designing and validating a two-stage framework that synergizes the strengths of both. The first stage employs a domain-specific CV module with custom algorithms to perform precise, rule-based geometric reasoning tasks, delivering reliable quantitative measures for regulatory checks where general VLMs typically fall short. The second stage leverages an LLM for its advanced natural language understanding, translating these structured findings into actionable insights and facilitating interactive user dialogue. Collectively, the study pursues three objectives: 1) to design a hybrid CV–LLM framework, 2) to implement a mobile-based intelligent assistant for hazard detection and safety dialogue, and 3) to validate the approach in real-world construction scenarios.

This paper first reviews existing literature and technical foundations relevant to computer vision and language models in construction safety. It then presents the methodological framework and development of the proposed intelligent assistant, detailing the object detection and interaction modules along with the mobile application interface. The experimental setup is subsequently introduced, followed by the evaluation procedures used to validate the accuracy of generated safety captions and assess the overall system effectiveness. The manuscript concludes with a discussion of key findings, limitations, practical implications, and directions for future research.

## 2. Literature review

This section provides an overview of prior research on computational approaches for construction site monitoring, focusing on methods relevant to hazard detection, safety compliance, and scene understanding. The review highlights the capabilities, limitations, and gaps in existing computer vision (CV) and vision-language approaches, setting the context for the development of the hybrid reasoning framework presented in this work.

### 2.1. Computer vision techniques in construction safety inspections

CV-based methods have emerged as promising solutions for monitoring various aspects of construction sites, including safety [11,12], progress [13], structural health [14], and productivity [15]. These techniques, encompassing algorithms such as classification, object detection, object tracking, and instance segmentation, are extensively applied for tracking workers and equipment, identifying hazards, preventing accidents, and classifying behavior. Object detection identifies and categorizes objects within images [16], while object tracking locates objects across video frames [5]. Instance segmentation assigns each pixel of an image to predefined classes while distinguishing similar objects. These algorithms are integrated into monitoring systems to provide real-time data for safety personnel.

CV-based detection algorithms are categorized into one-stage detectors, which directly detect objects from the full image, and two-stage detectors, which generate region proposals before classifying objects. While one-stage detectors are faster, they may be less accurate, particularly for small objects or cluttered scenes. Examples of two-stage detectors include Region-based Convolutional Neural Networks (R-CNN), Region-based Fully Convolutional Networks (R-FCN), and Mask R-CNN, while popular one-stage detectors include You Only Look Once (YOLO) series, Single Shot Detector (SSD), and RetinaNet. Researchers have employed two-stage detectors for object detection and instance segmentation in construction site safety, achieving high accuracy in tracking workers despite occlusions and scale variations [17–19]. For instance, studies such as those by Wang et al., (2019) and Wu et al., (2019) have demonstrated the effectiveness of Mask R-CNN in identifying and segmenting without-helmet behavior from surveillance videos of construction sites [18,20]. One-stage detectors are commonly used for real-world data, with studies focusing on improving performance under different weather conditions and customizing algorithms for specific safety monitoring applications [21,22]. Wang et al. (2021) and Kim et al., (2019) have utilized YOLO for object detection tasks in construction site safety, showcasing its capability to detect objects efficiently in different environments [23,24]. Similarly, Chern et al., (2023) proposed a vision-based modularized system for context-aware safety monitoring of workers to assess personal protective equipment compliance [25]. These systems used object detection, semantic segmentation, and depth estimation models to differentiate workers at height and on the ground and apply different PPE compliance rules.

Efforts to automate safety monitoring and inspection processes have centered on real-time visual data collection. It has been mainly used in semantic segmentation to discern and categorize all objects within images [26]. Similarly, the classification of site images has been pursued through the correlation of categorized scenes with specific safety rules [27]. For instance, Anjum et al., (2022) initially detected objects and then correlated their heights to classify the scene in accordance with safety protocols [28]. However, these methodologies are constrained by a basic limitation associated with the data capture process and its final representation. For instance, Liu et al., (2020) emphasized that scene elements are often examined separately, leading to a final representation typically composed of a predetermined combination of semantic labels presented as singular words or phrases [29]. A need is identified to find an automated methodology capable of analyzing construction site scenes, extracting diverse objects, connections between the objects, and

pertinent attributes from images in real-time, and converting them into comprehensive narratives, which can further be utilized as point of departure for safety discussions. This type of scene analysis within the construction domain has received relatively little attention. The subsequent section will delve into an in-depth exploration of the various methodologies for scene analysis presented by different research studies, as well as the associated gaps and limitations.

2.2. Methods for extracting and utilizing information from construction scenes

Advance applications for construction safety management rely solely on either vision-based or text-based data [30]. In recent years, a few efforts have been made to address this gap by adopting various methods and techniques (Table 1). Liu et al. (2020) proposes an image captioning method using two Deep Neural Network (DNN) compositions (i.e., VGG-16 with LSTM, and ResNet-50 with Long Short-Term Memory, LSTM) for scene description of three construction activities [29]. This approach involves linguistic descriptive schema to generate automatic image captions of construction images by utilizing CV and Natural language generation (NLG). While the results for the dedicated case scenarios presented in this paper are promising, the authors acknowledge significant limitations. This approach struggles to effectively capture the diverse nature of construction sites and practices. Additionally, the creation of caption datasets is costly and requires constant supplementation to expand sample size and categories, posing practical challenges. Moreover, it lacks the capability to interpret construction safety regulations corresponding to the scene.

To address these limitations, Wu et al., 2021 [31] and Fang et al., 2023 [32] introduced an approach that combines CV with ontology techniques to semantically reason about hazards and their mitigations by generating auto-phrases based on the given images. This method utilizes pre-defined Semantic Web Rule Language (SWRL) rules to formally represent safety regulatory knowledge alongside detected hazards and their mitigations. Ontologies excel at managing well-defined knowledge bases with a small set of entities and relationships. However, construction safety regulations can be intricate and evolving due to the dynamic nature of industry. When dealing with highly complex and dynamic knowledge structures, ontologies can struggle to scale

effectively [33]. Additionally, traditional ontology processing relies on rigid semantic engines, which may limit flexibility and adaptability. Storing a comprehensive safety knowledge base within an ontology can require significant storage space, posing scalability challenges for large-scale deployments [34]. Furthermore, encoding knowledge of numerous safety regulations into SWRL rules is time-consuming, which presents a significant challenge for the practical application of the system.

[7] and [29] proposed a different approach that generates spatial relationship construction scene graphs by utilizing computer vision (CV)-based descriptions, encompassing entities, attributes, and their interactions. This approach then employs a C-BERT Bidirectional Encoder Representations from Transformers, (Natural Language Processing (NLP)) network to reason about hazards by integrating these scene graphs with construction regulations. This integration allows the system to leverage the rich contextual information encoded in the scene graph and the regulatory knowledge embedded within the construction rules for improved hazard identification. However, the system faces limitations in practical applications. Scene graphs for relation detection heavily rely on labeled datasets, which necessitate a higher cost of dataset annotation with relations compared to the annotation of entities. Particularly for construction scenes involving multiple workers and machines, defining interrelationships can be exceedingly complex. Moreover, this approach lacks flexibility for dynamic rule generation, making it less adaptable to evolving construction scenarios.

Another approach involves comparing image captions with safety regulations by using dense image captioning techniques. Wang et al. (2023) [8] presented a vision-assisted behavior-based method, proposing a visual-text semantic similarity method. This enables the system to automate the safety hazard identification process using computer vision and facilitate reasoning about hazards in the caption by utilizing natural language processing. However, this approach can be limited by factors such as rare view angles, confusing construction materials, and mismatches between image content and captions. Additionally, the model may struggle to handle situations with multiple entities or may not be fully interpretable [35], particularly regarding the CNN-based encoder used for image processing. Furthermore, this method does not offer precise regulatory reasoning, which is crucial for safety compliance in construction sites.

The methods discussed in this section extract information from

Table 1  
Comparison of methods for utilizing visual data in the construction industry.

| Author and Year:          | [7]                                                                                 | [29]                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Technique and Objectives: | Scene Graph → BERT for Regulations checking                                         | Captioning model → Image Caption for scene description                                 |
| Example Image:            |  |    |
| Results:                  | Worker is unsafe, Falling, combating.                                               | A man wearing a yellow helmet and a yellow reflective vest is fixing the ceramic tile. |
| Limitations:              | Lacks natural language understanding for regulatory checks.                         | Lacks precision in detecting safety violations.                                        |
| Author and Year:          | [31]                                                                                | [8]                                                                                    |
| Technique and Objectives: | Visual information → SWRL rule-based ontology for description of hazards            | Dense Captioning → Semantic similarity for Regulation checking                         |
| Example Image:            |  |    |
| Results:                  | There are two types of hazards in image: falling from height and head injuries      | worker should wear a safety helmet," "worker should wear safety boots                  |
| Limitations:              | Lacks flexibility for dynamic rule generation.                                      | Requires large datasets, limiting scalability.                                         |

construction scenes by generating textual descriptions (sets of words or single phrases) based on visual data captured by cameras. However, these approaches face several limitations. As discussed, the reliability of these descriptions can be hampered by factors like dataset limitations or complex scene elements. Furthermore, data utilization is restricted, as it typically doesn't allow for user interaction or discussion about the generated content. This highlights the inability of these systems to bridge the gap between raw visual data and a comprehensive understanding of its safety implications. A recent study adopted a real-time image captioning technique for situational awareness and introduced a query system based on content for workers in augmented reality [36]. Nevertheless, this approach is limited by its reliance on textual data for caption generation, hindering their ability to precisely assess specific situations for inspection purposes in accordance with construction safety rules.

While recent advancements in vision-language models, such as GPT-4 V, BLIP-2, and MiniGPT-4, have significantly expanded the capabilities of AI in multimodal tasks, they remain largely focused on semantic-level understanding rather than precise geometric or quantitative reasoning [10]. These models can describe what is seen in an image and generate natural language explanations, but they lack the capacity for fine-grained spatial measurements or compliance verification grounded in safety standards. Although GPT-4 V exhibits geometric reasoning in tasks like GPS localization, their performance in construction safety monitoring is limited due to multi-entity scenarios, regulation-based evaluations, and real-time decision-making requirements. Additionally, existing literature often overlooks the need for affordable and accessible tools for safety managers to utilize the generated data effectively. To address these limitations and generate more accurate scene descriptions, this paper leverages Large Language Models (LLMs) for two key purposes:

- **Generating Scene Captions:** LLMs analyze computer vision (CV) data to create captions describing the construction scene.
- **Facilitating Interactive Discussions:** An interactive chatbot powered by LLMs can be accessible on users' mobile phones, enabling them to discuss potential solutions for identified hazards.

To mitigate potential limitations related to domain knowledge, this paper utilized prompt engineering techniques within the LLM framework. The next section will delve into the details of our research methodology, explaining how these methods are developed.

### 3. Research method

This section describes the development and evaluation process of the proposed system that integrates computer vision (CV) and language model processing (LLM) for analyzing construction scenes. The identification of the problem stemmed from a comprehensive review of the literature, followed by the proposal and evaluation of solutions for each identified issue.

The system's development phase commenced with the creation of a CV-based model, achieved through data collection and preprocessing techniques. Leveraging image data from publicly available datasets and augmenting it with additional images to ensure diversity in the training data. In the post-processing phase, safety rules relevant to objects were integrated to classify scenes in compliance with safety regulations. This process was adopted for three distinct scenarios.

Textual information derived from this process underwent LLM processing to generate descriptive captions, laying the foundation for the subsequent interactive chatbot module. The integration of these modules culminated in the development of an Android mobile application, tailored for ease of use by construction personnel on-site.

Evaluation metrics were employed to assess the accuracy of the object detection model, post-processing outcomes, and generated captions. User feedback, gauged using the semantic differential scale, provided

insights into the system's applicability and potential adoption. The paper's sequential workflow is illustrated in Fig. 1, elucidating the step-by-step progression of the paper.

#### 3.1. System development and integration

Fig. 2 illustrates the workflow of a construction safety analysis system accessible through a mobile application. An inspector captures a scene image, which is then processed by a computer vision module to detect objects and elements like workers, equipment, and safety gear. The system checks pre-defined safety rules associated with these detections to classify the scene as safe or unsafe. Information extracted from the image, including objects and the safety classification, is then used as a prompt for a Large Language Model (LLM). This LLM generates a descriptive caption summarizing the scene. Finally, an interactive chatbot within the application presents the caption to the user, allowing for further discussion, questions, or additional information.

For instance, the system might analyze an image containing an A-type ladder scenario with multiple safety violations. By checking Occupational Safety and Health Administration (OSHA) and Korea Occupational Safety and Health Agency (KHOSA) guidelines, the system would identify violations such as workers without helmets, workers positioned at excessive heights, and workers standing on prohibited ladder positions. In this A-type ladder case, the system would then classify the scene as unsafe and generate a comprehensive caption like: "Unsafe Behavior: A worker is working at a height of 1.82 meters (greater than 1.2 meters) and at the top of a ladder. Although a co-worker is present at the top of the ladder, this configuration is risky for both workers according to OSHA. Additionally, the absence of helmet and improper use of a ladder without an outrigger are also unsafe." The ladder used for this study is only A-type ladder. The inspector could then use the chatbot interface to discuss potential solutions. This information can empower the inspector to address the identified safety hazard and make informed decisions within the constraints of resource

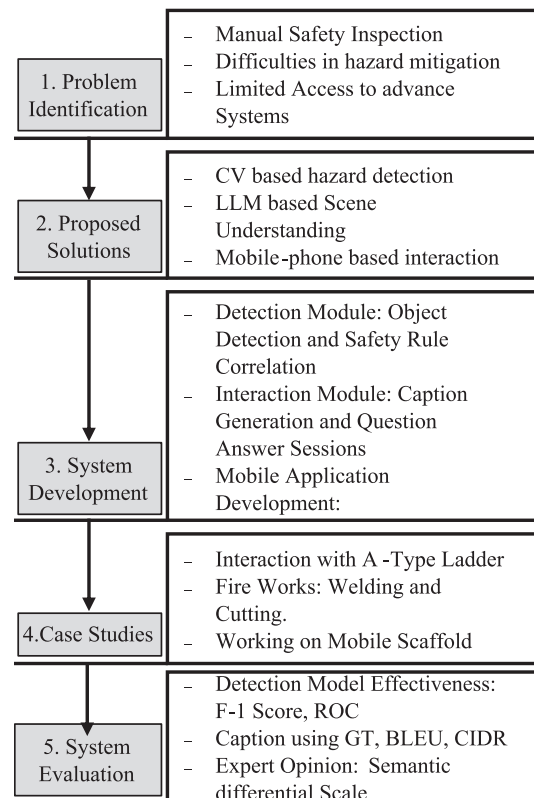


Fig. 1. Research method.



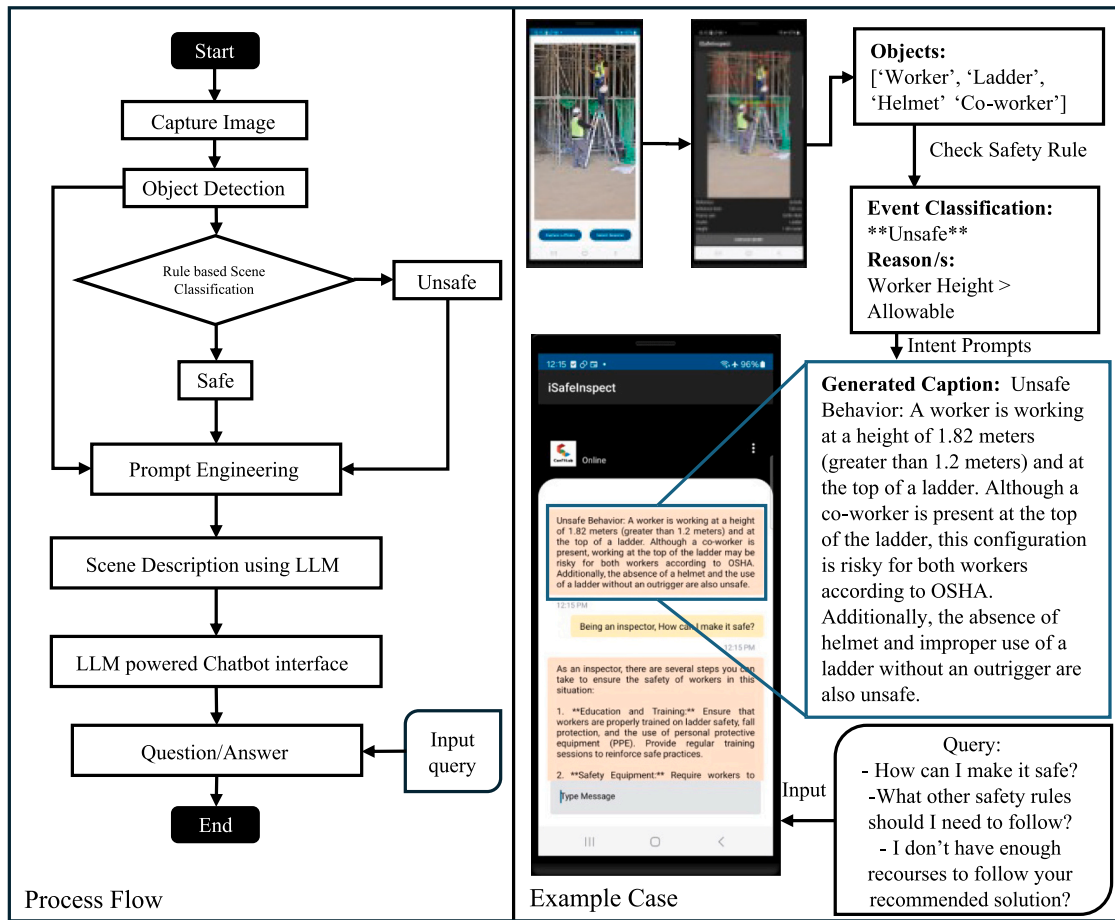


Fig. 2. Proposed system process flow with example.

availability. Further the chatbot can assist inspectors with solutions to improve safety, for instance, (How can I make it safe?), explore relevant safety regulations (What other safety rules should I follow?), or even express limitations in implementing suggested safety measures ("I don't have enough resources to follow your recommended solution?").

The detection module begins the first phase from the application layer of System Architect (Fig. 3). This module employs advanced computer vision algorithms to identify objects within images, assigning unique identifiers and classifying them accordingly. This module also computes spatial coordinates for the detected objects, facilitating their subsequent analysis within the system. Furthermore, this module incorporates safety rules to classify scenarios as either safe or unsafe events. Additionally, this layer integrates a rule-based safety analysis, categorizing observed scenarios as safe or unsafe, with the rationale for this determination being articulated through sophisticated post-processing techniques. The interaction module, constituting the second phase of the application layer, utilizes visual descriptors such as object information, event status, and causative factors to construct structured prompts. These prompts act as inputs as a caption generator, which then interfaces with LLMs to produce contextually relevant captions as well as the predicted hazards in the scene. These captions not only provide descriptive information but also initiate further interaction with the scene.

The database layer, located at the system's core, serves as a repository for storing original images, detection results, and generated captions. This layer ensures efficient data management and retrieval for ongoing and future analytical needs. Stored information can be accessed to facilitate further discussions about captured scenarios. To initiate a conversation, a structured prompt containing information from the database is provided, along with an instruction to enter conversational

mode. This conversational mode can be accessed via the mobile application interface, allowing users to pose specific questions about detected scenes and receive responses from the system. Leveraging the extensive knowledge base of LLMs, users can engage in interactive dialogue, enhancing their understanding of the captured scenarios.

### 3.1.1. Detection module: object detection and safety rule correlation

Identifying hazards and checking safety rules from computer vision techniques usually require two steps, first identifying objects, entities and then finding correlation between them to check the compliance of safety rules [37]. The initial step involves rigorous preprocessing of the image dataset to optimize it for training the object detection model (Fig. 4). This includes various transformations such as resizing the images to a standard input size, often chosen based on the architecture requirements, to ensure uniformity in input dimensions across the dataset. Furthermore, augmentation techniques are employed to enrich the dataset and improve the model's ability to generalize to diverse scenarios. Augmentation encompasses a spectrum of transformations, including rotation, scaling, translation, and changes in brightness and contrast, applied randomly to each image during training.

Following dataset preparation, the YOLOv12 state-of-the-art object detection framework was employed to train the model [38]. YOLOv12 introduces a transformative attention-centric architecture that departs from traditional CNN-based designs. Central to its innovation is the Area Attention module, which segments feature maps into horizontal or vertical stripes, enabling efficient self-attention across large receptive fields while maintaining computational efficiency. Complementing this is the Residual Efficient Layer Aggregation Network (R-ELAN), an enhanced backbone that facilitates deeper feature integration through residual connections, improving gradient flow and feature reuse.

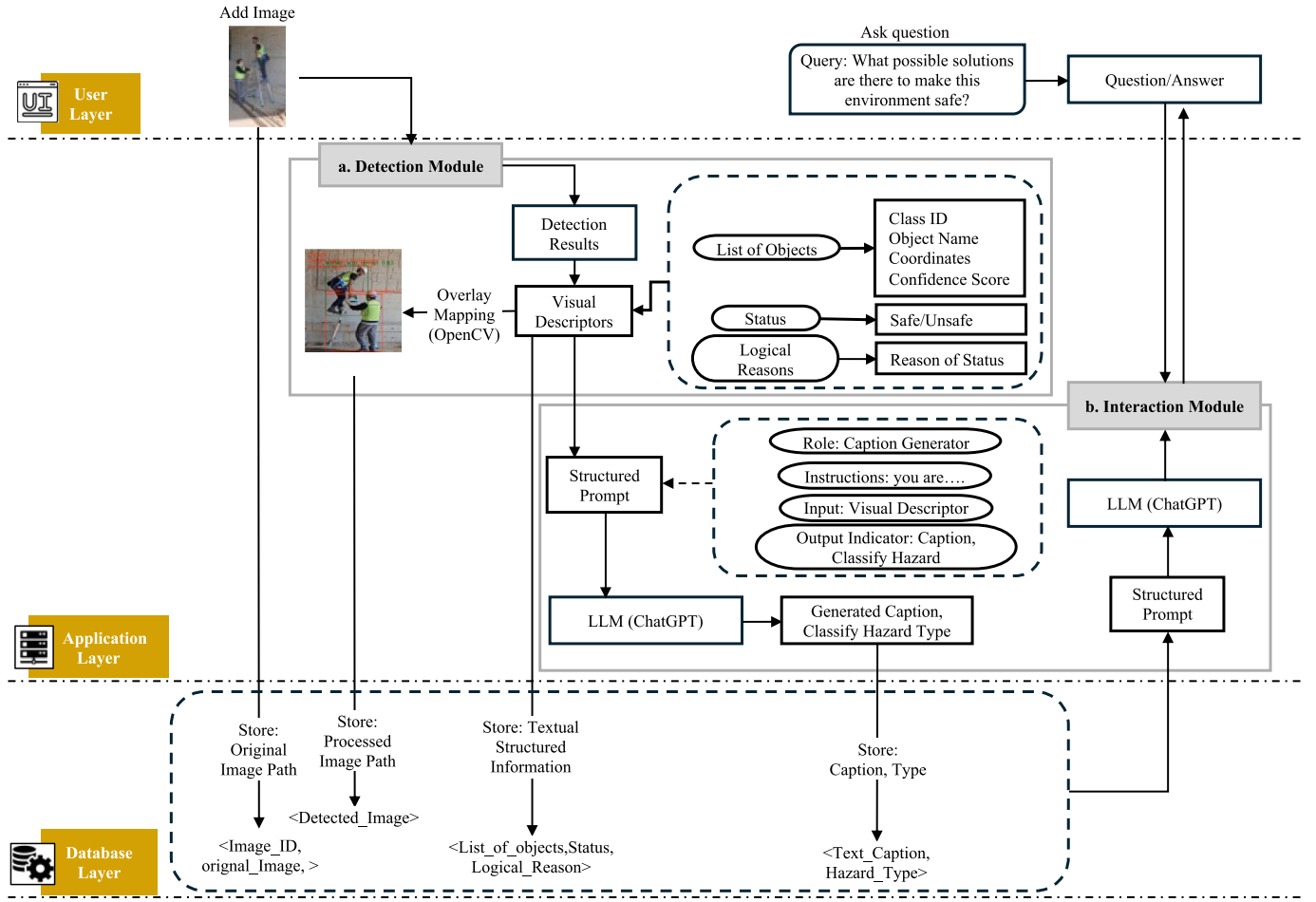


Fig. 3. Proposed system architecture.

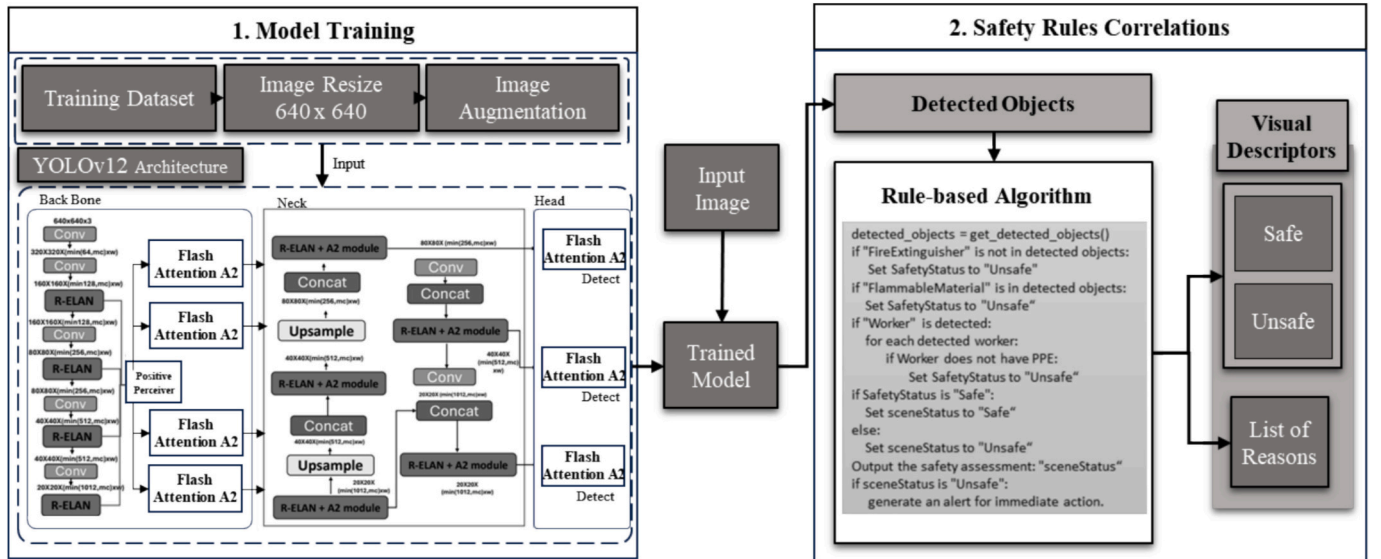


Fig. 4. CV-based detection and safety rule correlations.

Additionally, YOLOv12 employs  $7 \times 7$  separable convolutions to capture broader contextual information with reduced parameters, and integrates Flash Attention to accelerate attention computations, thereby enhancing both speed and accuracy in real-time object detection tasks.

Upon training the computer vision model on the preprocessed

dataset, the resultant model is adept at identifying objects within images swiftly and accurately. During the inference phase, an input image undergoes a series of transformations and passes through the trained model, yielding a set of bounding boxes encompassing the detected objects. Each bounding box is associated with a confidence score,

indicating the model's certainty regarding the presence of the corresponding object. Object detection provides valuable bounding box coordinates for identified objects, yet it alone is insufficient for assessing whether a person is actively engaged with the object under consideration or determining the number of people present on it at any given time. While the initial object detection process provides bounding box coordinates for identified objects, additional post-processing techniques are indispensable to accurately ascertain worker behavior during tasks and ensure adherence to safety protocols.

The post-processing stage involves analyzing the bounding box coordinates provided by the object detection algorithm and further interpreting the interactions between workers and objects. By analyzing the spatial relationships and interactions between workers and equipment, post-processing techniques enable the system to infer the nature of worker activities and assess safety compliance. Therefore, these techniques involve correlating the detected objects with safety regulations through a rule-based algorithm. This algorithm evaluates the presence or absence of specific objects, such as safety gear like helmets or vests, and compares them against predefined safety rules. For instance, if a person is detected wearing a helmet, the algorithm classifies the scene as safe; otherwise, it may flag it as unsafe. Visual descriptors are then generated, encapsulating pertinent information such as the classification of safety (safe or unsafe) and a rationale for the classification decision. The visual descriptors serve as inputs for the subsequent module, which focuses on generating captions for input images. Leveraging the safety classifications and rationales provided by the visual descriptors, the next module produces descriptive text that summarizes the content of the input image while also classifying the image based on relevant safety rules identified by the system.

### 3.1.2. Interaction module: caption generation and question answer sessions

The interaction module in this system plays a critical role in bridging the gap between visual scene analysis and meaningful textual descriptions, ultimately enabling real-time, context-sensitive discussions between users and the system. This module integrates computer vision outputs with a Large Language Model (LLM) to generate accurate and relevant image captions and to support interactive question-and-answer sessions, fostering a more informed understanding of construction site safety.

As illustrated in Fig. 5, the process begins in the left panel, where

visual descriptors derived from the computer vision component are processed into structured prompts. These prompts serve as inputs to the LLM's transformer-based encoder-decoder architecture, shown in the center panel. The encoder takes these prompts and encodes them into dense, context-rich vector representations using multiple layers of self-attention and feed-forward networks. This encoding process is crucial for capturing the intricate dependencies between different elements of the scene, such as the proximity of workers to hazardous equipment or the absence of required personal protective equipment (PPE). In the right panel, the decoder leverages these encoded vectors to generate coherent, context-aware image captions and interactive question-answer outputs. The decoder employs masked multi-head attention mechanisms to predict the next word in the caption sequence while considering all possible previous words. This iterative prediction process continues until the caption is fully generated. The resulting caption not only summarizes the scene but also highlights any identified safety violations, providing the user with an immediate and concise description of the safety status.

Beyond caption generation, the interaction module extends its functionality by enabling real-time, interactive dialogues with users (Right Panel of Fig. 5). The generated caption, along with other relevant contextual information, is used to initiate a conversation through the system's chatbot interface. This interactive component allows users to ask follow-up questions or seek further clarification about the safety assessment. For example, the chatbot can respond to queries like "What specific rule is violated in this scene?" by retrieving and presenting the relevant safety regulation and its context. This fosters a deeper understanding of the safety issues present and supports informed decision-making.

Additionally, the chatbot supports various user interactions, including requests for mitigation strategies, explanations of applicable safety regulations, and discussions of resource constraints. Users can inquire about potential solutions to rectify unsafe conditions, such as asking, "How can we make this area safer?" The chatbot responds by suggesting specific actions aligned with best practices, such as installing guardrails or ensuring proper PPE use. The system can also handle questions like "What other safety precautions should be taken in this situation?" by generating recommendations that comply with industry standards. Moreover, in situations where resource limitations are a concern, the chatbot can propose alternative measures that are feasible

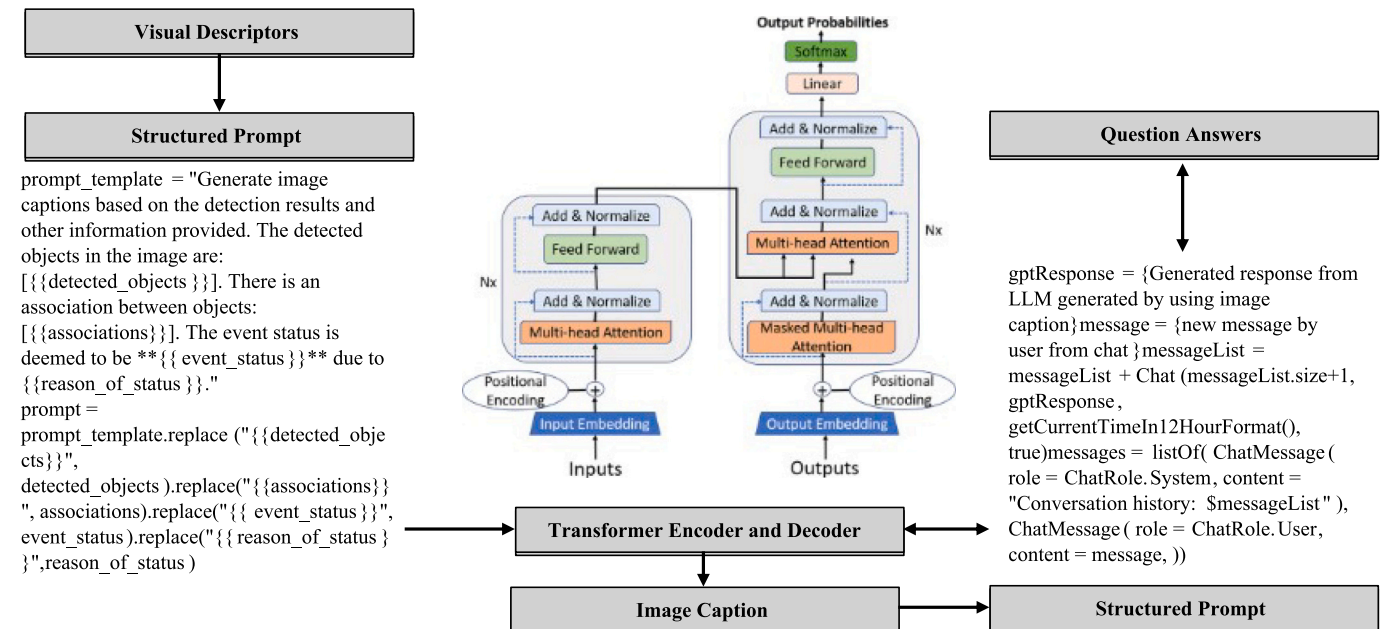


Fig. 5. System architecture for interaction module.

within the constraints mentioned by the user, such as optimizing the use of existing resources or prioritizing the most critical safety interventions. The LLM-powered chatbot facilitates a continuous learning process for safety inspectors, helping them address specific hazards while also broadening their understanding of general safety practices. The interaction module, therefore, not only provides immediate insights into site safety but also empowers users to engage actively in discussions about safety management, ultimately contributing to a safer work environment.

### 3.1.3. Mobile application development

The descriptive scene classification mobile application, developed in Android Studio LGUANA (2023.2.1 Patch 2), is designed for Android smartphones running SDK versions 21 to 33, covering approximately 99.5 % of Android devices that can operate the Application. The app requires the “Internet” permission at installation to enable API interactions with a large language model. It also seeks runtime permissions for accessing the camera and reading photos, media, and files from the device, essential for capturing images directly from the camera or accessing stored photos.

The mobile application comprises a series of five primary activities (Fig. 6), designed to optimize user experience and functionality. Upon application launch, users encounter the Login Activity, serving as the initial checkpoint for secure access. Robust authentication mechanisms, such as Firebase Authentication, are employed to ensure stringent access control and data security. Following successful authentication, users transition to the Capture Scene Activity, where the CameraX API and Android’s media store API are leveraged for real-time scene capture. This gives users the option to either load an image from the gallery or capture one using the camera. Integration of advanced image processing libraries like Glide are used to enhance functionality by ensuring efficient image loading and transformation.

The subsequent activity entails the selection of detectors, wherein users are presented with a predefined list of case scenarios. In this paper, these scenarios encompass the assessment of 1) Use of Ladder, 2) Use of Mobile Scaffold, and 3) Checking the Fireworks for Cutting and Welding. Leveraging dynamic loading techniques, users are empowered to customize the subsequent detection process to align with their specific needs and requirements. Upon selecting a scenario, users are guided to the pivotal Detection Activity, where computer vision technology acts as a central role in the application’s functionality. During this stage, the system utilizes the TensorFlow Lite GPU library (version 2.8.0),

integrated via Gradle, to showcase its advanced object detection capabilities. Images are processed through machine learning algorithms using the TensorFlow Lite interpreter, allowing for the classification of scenes as safe or unsafe. Moreover, users are provided with a comprehensive list of detected objects, complete with class IDs, bounding box coordinates, and confidence scores. In instances of unsafe scenes, users receive detailed explanations, fostering heightened situational awareness and informed decision-making. Lastly, the Chat Activity exemplifies the application’s integration with natural language processing (NLP) technologies. Leveraging the LLM API from OpenAI (client version 2.7.1), users engage in intuitive text-based interactions. The chat function retains the complete chat history throughout the session, ensuring that each response remains consistent with both the previously generated caption and the ongoing conversation context. The implementation includes a recycler view for chat message display, with a robust model class managing data efficiently. Additionally, a local SQLite database, implemented using Android’s Room Persistence Library, stores chat data, ensuring seamless continuity and effortless reference of past interactions.

The UI components of the application are defined using XML (extensible Markup Language), which structures the visual elements, while the logical operations are handled in Kotlin files. Kotlin, developed by JetBrains, is a concise and expressive programming language for Android app development, featuring null safety and coroutines for asynchronous tasks. This clear separation aids in the efficient development and maintenance of the application, with XML managing the interface layout and Kotlin implementing the functional aspects. Table 2 provides a concise summary of each mobile application activity, detailing the activity name, primary function, key APIs and libraries used, and the resulting outputs to complement the descriptive text.

### 3.2. Experiment, evaluation and results

In this section, three different scenarios inspired by existing literature were adopted to evaluate the performance of the detection module. Following the integration of the object detection algorithm and the large language model (LLM), descriptive safety-related captions were generated based on the identified hazards. Finally, user satisfaction with the overall system was assessed. The comprehensive evaluation framework is illustrated in Fig. 7. As shown, the system validation encompasses three core components: (1) the detection module, (2) caption generation, and (3) user satisfaction. The detailed experimental procedure,

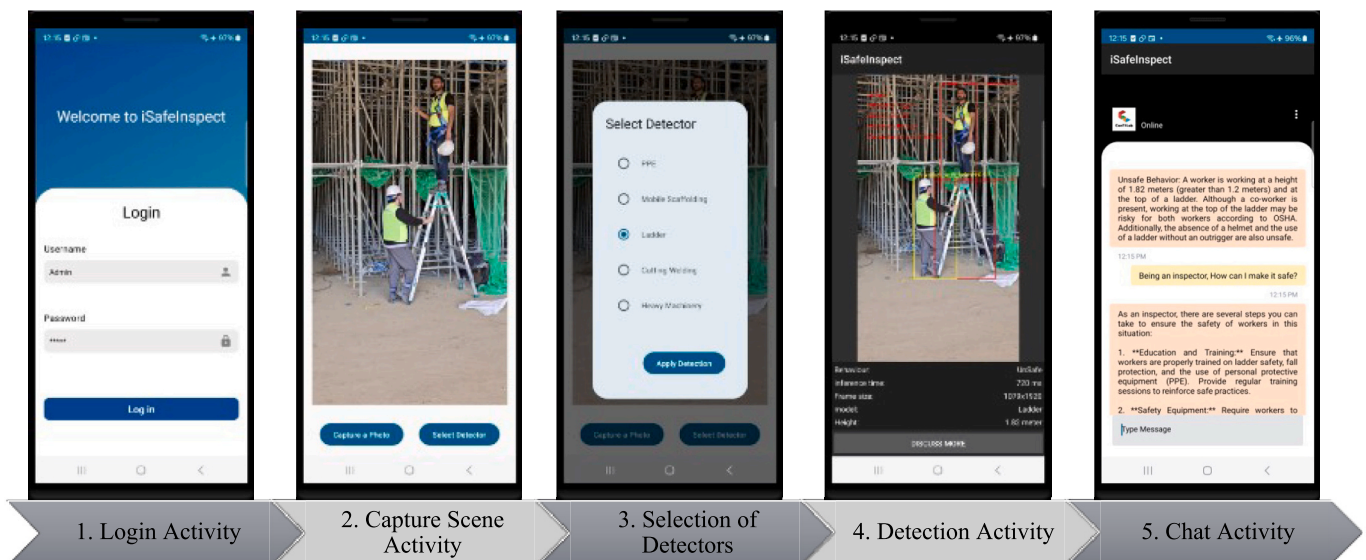


Fig. 6. User interface flow for mobile application.



**Table 2**  
Mobile application activities summary.

| Activity Name          | Function                                      | Key APIs/Libraries                                               | Output                                                                                    |
|------------------------|-----------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Login Activity         | Secure user authentication                    | Firebase Authentication SDK                                      | Authenticated user session                                                                |
| Capture Scene Activity | Image acquisition and preprocessing           | CameraX API, Android Media Store API, Glide v4.14.2              | Preprocessed image for analysis                                                           |
| Selection of Detectors | Scenario-specific detector configuration      | Dynamic UI components                                            | Selected detection scenario                                                               |
| Detection Activity     | Object detection and safety classification    | TensorFlow Lite GPU v2.8.0, TensorFlow Lite Interpreter          | Detected objects (class IDs, bounding boxes, confidence scores) and safety classification |
| Chat Activity          | Natural language interaction and consultation | OpenAI API Client v2.7.1, Room Persistence Library, RecyclerView | Interactive chat responses with stored conversation history                               |

evaluation metrics and results for each component are detailed in the subsequent subsections.

### 3.2.1. Detection module

For performance evaluation of the computer vision-based detection module of the proposed system, three different construction scenarios were considered, namely, inspection of ladder, mobile scaffold, and welding task usages. These scenarios were selected due to their prevalence and inherent hazards within the construction industry. To achieve this, the existing datasets from previous studies were extended with a diverse range of images for each scenario, as outlined in Table 3. This integration aimed to enhance the accuracy and robustness of the proposed safety monitoring system. The three work activities included are the use of ladders, use of mobile scaffolds, and firework activities (welding and cutting), each accompanied by specific safety regulations derived from Occupational Safety and Health Administration (OSHA) and Korea Occupational Safety and Health Agency (KHOSA) guidelines.

For ladder usage, the dataset comprises 4530 images that reflect safety-critical elements such as workers, helmets, ladders, and outriggers. The associated regulations emphasize proper equipment condition, the mandatory use of outriggers, adherence to allowable working heights, and helmet usage. Similarly, the mobile scaffold scenario includes 4868 images that capture relevant components like guardrails and outriggers. The regulations mandate the installation of guardrails, the use of helmets, and restrict the number of workers on scaffolds to two at a time to prevent overloading. The firework scenario includes 2070 images and focuses on safety measures during welding and cutting activities, such as ensuring the presence of fire extinguishers and prevention nets and checking for flammable materials.

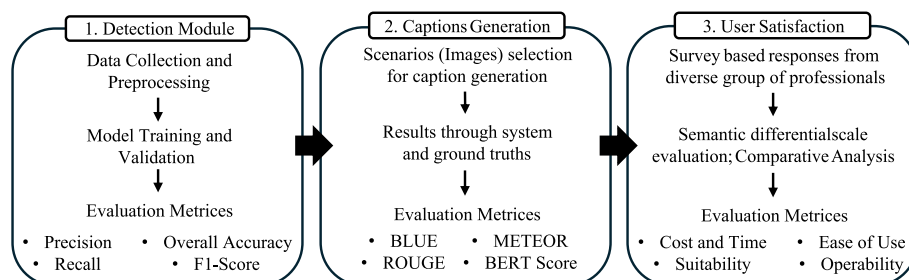
All datasets were divided into training, validation, and testing subsets, as presented in Table 4. The training subset was used for model learning, while the validation set was employed for hyperparameter optimization and monitoring potential overfitting. The final

performance of the models was evaluated on the independent test set. For the classification module, the images from the test set were manually labeled as safe or unsafe according to safety rules to establish ground truth. A high-performance workstation was used for training and testing the object detection models as well as for evaluating the classification module. The workstation was equipped with an Intel Core i9–12,900 CPU @ 2.4 GHz, 32 GB RAM, and an NVIDIA RTX 3090Ti GPU with 24 GB VRAM. To assess practical usability in real-world scenarios, the proposed framework was also deployed on a Samsung Galaxy S23 Ultra smartphone equipped with 8 GB RAM and Android OS version 14.

To mitigate the influence of image quality variations on detection reliability, the system incorporates several control mechanisms. First, during model training, extensive data augmentation was applied (including adjustments in brightness, contrast, and sharpness, as well as random rotations and blurring) to simulate diverse real-world conditions. This enhanced the model's ability to generalize images of varying quality. Second, the mobile application integrates automatic pre-processing filters that normalize brightness and contrast before sending an image to the detection module.

**Table 3**  
Safety requirements and corresponding regulations.

| Construction Entity | Safety Requirement                      | OSHA [39]                                    | KOSHA [40]                   |
|---------------------|-----------------------------------------|----------------------------------------------|------------------------------|
| A-Type Ladder       | Used with outriggers                    | –                                            | Article 68 (1), [C-28-2018]  |
|                     | Maximum working height 1.2 m, co-worker | –                                            | Article 68 (3)               |
|                     | Prohibition of workers on ladder top    | OSHA 1926.1053(b) (13)                       | Article 68 (3)               |
|                     | Helmet required                         | OSHA 1926.100 (General Requirements)         | Article 52                   |
| Mobile Scaffold     | Outriggers required                     | OSHA 1926.452(w)(6) (iii)                    | Article 68 (1), [C-28-2018]  |
|                     | Guardrail required                      | OSHA 29 CFR 1926.451(g)(4)(i)                | Article 68 (1)               |
|                     | No more than two workers                | 1926 Subpart L, Appendix A, Section 2 (g)(4) | Article [C-28-2018]          |
|                     | Worker helmet required                  | OSHA 1926.100 (General Requirements)         | Article 52                   |
| Welding/Fire Work   | Fire extinguisher present               | OSHA 1926.352(d)                             | Article 282, Article 80-(20) |
|                     | Flammable materials present             | OSHA 1926.352(c)                             | Article 282                  |
|                     | Fire prevention net installed           | –                                            | Article 282                  |
|                     | Hard hat required                       | OSHA 1926.100 (General Requirements)         | Article 52                   |



**Fig. 7.** Evaluation strategy and process flow.

**Table 4**

Dataset description and classification.

| Work Scenario                                                              | Description                                                                                           | Classes                                                                                                                  | Dataset<br>(Original +<br>newly added)                                 |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Use of Ladder<br>[28]<br>See Algorithm<br>1 (Appendix 1)                   | For ladder safety, ensure each component is in good condition and adhere to regulatory guidelines.    | 1.Worker<br>2.Helmet<br>3. Ladder<br>4. Outriggers                                                                       | 1825 + 2745 = 4570 images<br>Train = 3653<br>Valid = 460<br>Test = 457 |
| Use of Mobile Scaffold [26]<br>See Algorithm<br>2 (Appendix 1)             | Ensure safe usage of mobile scaffolds by inspecting components and following regulatory requirements. | 1. Worker<br>2. Helmet<br>3. MScaffold<br>4.Outriggers<br>5. Guardrails                                                  | 933 + 3935 = 4868 images<br>Train = 3902<br>Valid = 483<br>Test = 483  |
| Fire Work<br>(Welding and Cutting) [41]<br>See Algorithm<br>3 (Appendix 1) | Implement safety measures for welding tasks, including equipment checks and adherence to regulations. | 1.Welding Equipment,<br>2.Circular Saw<br>3.Fire Extinguisher<br>4.Flammable Material<br>5.Prevention Net<br>6. Hard Hat | 633 + 47,235,356 images<br>Train = 4934<br>Valid = 222<br>Test = 200   |

Recently, YOLOv12 [38] by Ultralytics has emerged with notable improvements in accuracy and robustness, incorporating advanced feature extraction, anchor-free design, and an optimized backbone. Its single-pass architecture enables real-time detection under challenging conditions such as variable lighting and occlusion. Given these capabilities, YOLOv12 was employed as the underlying architecture for our computer vision system. Object detection provides bounding box coordinates for the detected objects, but it is insufficient to assess whether a person is working with object under consideration or to determine how many people are on it at any given time. Therefore, rule-checking-based algorithms were adopted and implemented to systematically evaluate adherence to safety rules. These algorithms are aimed at identifying unsafe behaviors or situations within each scenario. For example, when assessing compliance with the rule regarding worker height while on a ladder, the system first analyzes the coordinates of the worker and ladder bounding boxes. By establishing the relationship between these bounding boxes, the system determines if interaction between the worker and ladder is occurring. Subsequently, the vertical distance between the top of the worker's bounding box and the bottom of the ladder is calculated to determine the worker's height from the ground. If this height exceeds the specified limit, the system flags the result as unsafe, citing "worker exceeding height limit,".

The worker's height is estimated from the bounding box geometry of the detected worker and ladder, using a fixed ladder height of 2.0 m as the reference. The relative position of the worker's bounding box is proportionally scaled to this reference to obtain the estimated working height. This approach follows the same height assessment strategy introduced by Anjum et al. [28], where bounding box pixel values were normalized against a calibrated ladder height to compute worker elevation.

Algorithm 1 serves as the main controller function for safety detection across three construction scenarios. The algorithm performs scenario-specific model selection and delegates geometric safety analysis to specialized algorithms. For ladder safety (Algorithm 2-ML), it evaluates outriggers deployment compliance, worker height, and helmet usage. For mobile scaffolding (Algorithm 3-MMS), it assesses guardrail presence, outrigger deployment, and worker capacity limits. For

cutting/welding operations (Algorithm 4-MCW), it monitors PPE compliance and fire safety equipment availability. Each detection model (ML, MMS, MCW) represents scenario-optimized YOLOv12 architectures trained on specific construction safety elements, with the geometric measurement and compliance assessment performed by the corresponding specialized algorithms (Algorithms 2–4, detailed in Appendix 1).

These algorithms are evaluated using distinct performance indicators, such as precision, recall, precision-recall (PR) curve, and mean Average Precision (mAP), which offers a robust measure of the model's detection. Precision and recall evaluate the model's ability to correctly identify hazards and avoid false positives, while the PR curve demonstrates the model's effectiveness across varying decision thresholds [42]. mAP is especially critical for multi-class object detection evaluation, delivering a holistic measure of the model's detection accuracy across diverse object categories and varying confidence levels [43]. Since this framework also involves binary classification (safe vs. unsafe), we further employed F1-score, and accuracy receiver operating characteristic AUC-ROC, which are widely used for assessing classification effectiveness, especially under class imbalance [43].

Findings from the evaluation, demonstrated through PR and ROC curves along with classification results, present clear validation of the method's effectiveness across three distinct safety scenarios: ladder use (mAP = 97.9 %), mobile scaffolding (mAP = 97.5 %), and welding work (mAP = 98.9 %) as shown in Fig. 8. Results from these metrics underscore the reliability of the approach in addressing safety-related scenarios. The system's performance is detailed in Table 5, highlighting robust precision, recall, and F1-scores across diverse construction safety scenarios. For ladder-related tasks, the model achieved overall precision of 94.2 % and recall of 93.3 %, indicating strong capability in distinguishing between safe and unsafe behaviors. Notably, it classified safe behaviors with 94 % precision, 93 % recall, and 93 % F1-score. In the mobile scaffolding scenario, the model attained 98.7 % overall precision and 87.0 % recall, reflecting its high sensitivity in detecting hazards even under complex conditions. For safe behavior classification, it achieved 99 % precision, 87 % recall, and 93 % F1-score. For fire-related tasks such as welding and cutting, it achieved overall performance with 98 % precision and 91 % recall. Safe behavior classification demonstrated exceptional performance with 100 % precision, 95 % recall, and 97 % F1-score.

End-to-end latency for the object detection module is 80–100 ms per frame (10–12 FPS) on mobile devices. ChatGPT-based conversational responses incur 1.5–3.5 s latency but operate asynchronously to avoid interrupting real-time detection. This ensures immediate safety alerts while providing contextual guidance shortly after.

### 3.2.2. Validation of generated captions

To validate the captioning module of the proposed system, a comprehensive evaluation was conducted using ten diverse visual inputs, where each scenario represents a distinct construction safety situation and task (e.g., ladder use, mobile scaffolding, welding) captured in an image with varying environmental conditions, worker activities, and safety compliance states. These scenarios were systematically selected to cover a broad spectrum of real-world construction environments, ensuring comprehensive assessment of the model's performance across different safety contexts including ladder operations, scaffolding work, and fire/welding activities. Each scenario encompasses multiple variables: environmental conditions (lighting, weather, site layout), construction activities (different types of work being performed), safety equipment presence or absence (helmets, harnesses, outriggers), and regulatory compliance status (adherence to regulatory standards). For example, Fig. 9 illustrates one scenario with system generated caption involving ladder usage where the system identifies: 1) worker positioning relative to safety regulations (height of 1.82 m exceeding the 1.2-m threshold), 2) safety equipment assessment (absence of helmet and improper ladder setup without outriggers), and 3) regulatory

---

**Algorithm 1 Main function for safety detection**

---

Input: image\_path, model\_type, Detection Models [ML, MMS, MCW]  
Output: overall\_status, safety\_assessment, reasons

```

1: Set overall_status ← Safe
2: Initialize detection_results ← [ ]
{Step 1: Load the Image}
3: I ← Load image from image_path
{Step 2: Select the model ML, MMS, MCW, model_type }
4: if model_type = "ladder" then
5:   model ← ML
6:   R ← model(I)
7:   overall_status, IA, reasons ← detect_ladder(I, R) // Calls Algorithm 2
8: else if model_type = "mobile_scaffolding" then
9:   model ← MMS
10:  R ← model(I)
11:  overall_status, IA, reasons ← detect_mobile_scaffolding(I, R) // Calls Algorithm 3
12: else if model_type = "cutting_welding" then
13:  model ← MCW
14:  R ← model(I)
15:  overall_status, IA, reasons ← detect_cutting_welding(I, R) // Calls Algorithm 4
16: else
17:   Error: Unknown model type
18:   return Error
19: end if
{Step 3: Display Image with final status}
20: return overall_status, safety_assessment, reasons

```

---

compliance evaluation (violation of regulatory standards), demonstrating the multifaceted nature of each evaluation scenario.

The system autonomously generated textual descriptions for each input, detailing observed safety conditions, such as the presence or absence of protective equipment, potential hazards, and adherence to safety protocols. To establish robust evaluation benchmarks, the same visual scenarios were independently evaluated by two expert groups: (1) Safety Inspectors with practical field experience (Ground Truth 1), who assessed images based on regulatory standards and real-world safety practices, and (2) Researchers specializing in construction safety (Ground Truth 2), who evaluated scenarios against established theoretical guidelines and academic frameworks.

The validation was performed using a combination of widely recognized evaluation metrics—BLEU, ROUGE, METEOR, and BERT Score [44]. These metrics were chosen to quantitatively assess the alignment between the system-generated descriptions and the expert-provided ground truths. Each metric offers a unique perspective: BLEU measures n-gram precision, ROUGE focuses on recall by assessing overlapping textual elements, METEOR accounts for both precision and recall with additional synonym and stemming analysis, and BERT Score evaluates semantic similarity by leveraging contextual embeddings. The combination of these metrics provides a comprehensive view of the model's ability to produce accurate, lexically similar, and semantically meaningful captions [44,45].

Fig. 10 presents the distribution of evaluation scores across all scenarios, the box plot analysis reveals a clear performance hierarchy: BERTScore and ROUGE achieve superior performance (means of 0.442 and 0.412 respectively), while METEOR shows moderate performance (0.301) and BLEU demonstrates the lowest scores (0.064). Notably, BERTScore exhibits the most consistent performance across scenarios (smallest standard deviation of 0.086), indicating reliable semantic understanding capabilities. This variability reflects the diverse complexity of construction safety scenarios and suggests that the system performs particularly well on specific types of safety assessments while maintaining overall robust semantic understanding capabilities.

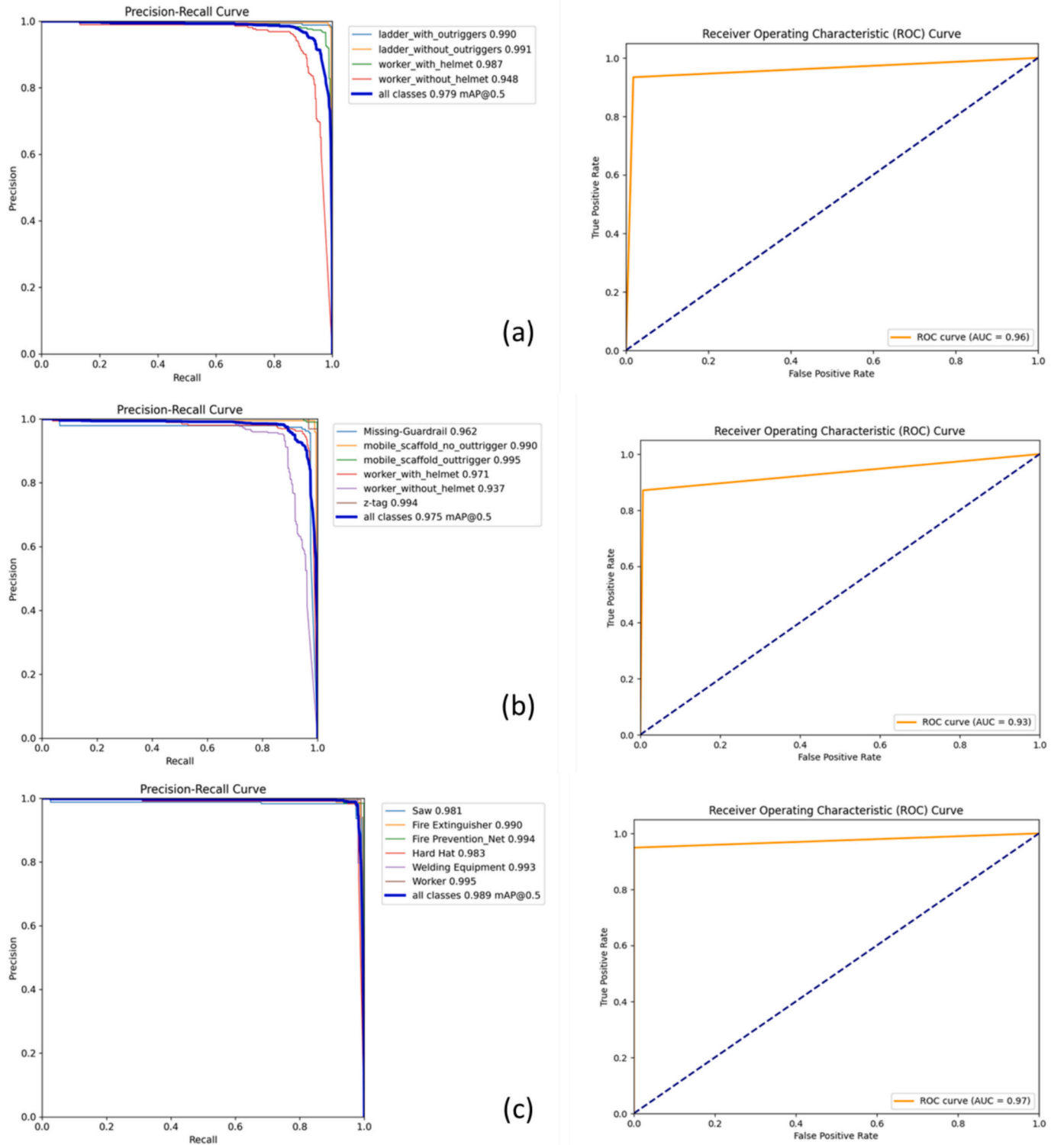
### 3.2.3. System effectiveness assessment

To evaluate the effectiveness of the system, user experience was assessed using the semantic differential scale. The evaluation involved a

diverse group of participants, including five safety inspectors (ages 28–42, with 7–10 years of experience, 60 % male and 40 % female), six construction management graduates (ages 28–35, with 3–7 years of experience, all male), and four researchers (ages 28–36, specializing in construction automation, with 1–2 years of experience, 75 % male and 25 % female). To ensure participants gained a nuanced understanding of the system's capabilities, they were introduced to its mobile application interfaces through detailed demonstration. Subsequently, participants engaged with the application across the designed construction scenarios, such as checking the safe use of ladder, mobile scaffolding, and welding tasks and asking questions through interaction module.

To quantify the user experience, a semantic differential scale consisting of bipolar adjectives across four critical criteria was employed [46,47]. Each criterion was scaled from 1 to 7 as presented in Table 6. The assessment criteria were systematically selected based on established frameworks for construction safety technology evaluation, focusing on both technical performance and visual design perspectives: 1) Time-efficiency and cost-effectiveness were evaluated following mobile application rating scale for e-tools assessment [48]; 2) Suitability for inspection operations was assessed using established construction safety inspection guidelines to determine regulatory compliance alignment [48]; 3) Ease of use was measured using semantic differential adaptations of NASA-TLX principles, validated for assessing user interface intuitiveness in complex operational environments [49]; and 4) Operability under dynamic conditions evaluated system reliability and stability under real-world construction site scenarios [49]. This multi-dimensional evaluation approach ensures comprehensive assessment across technical functionality, user interface design, regulatory compliance, and operational reliability dimensions essential for mobile construction safety technology deployment.

The user satisfaction level of the proposed system, as delineated through the statistical analysis provided in Table 6, highlights its significant effectiveness across crucial operational aspects. The evaluation reveals distinct performance patterns across different criteria: Participants rated the system's performance in reducing time and cost with mean scores of 6.33 and 5.87, respectively, indicating notable efficiency gains and cost savings. Moreover, from a managerial standpoint, the system's ability to streamline inspection processes bears implications for resource allocation and project timelines. Additionally, the system



**Fig. 8.** PR-Curve and ROC curve for (a) Activity 1- working on ladder, (b) Activity 2- working with mobile scaffold (c) Activity 3- fire works.

received favorable ratings for suitability and comprehensiveness, averaging 5.33 and 5.67, respectively, reflecting its adaptability to diverse inspection scenarios. This versatility not only enhances operational efficiency but also contributes to enhanced safety protocols, ensuring compliance with industry standards. Users also expressed satisfaction with the system's usability, with a mean score of 6.27, indicating intuitive design and user-friendly features.

Fig. 11 illustrates user satisfaction percentages across three distinct

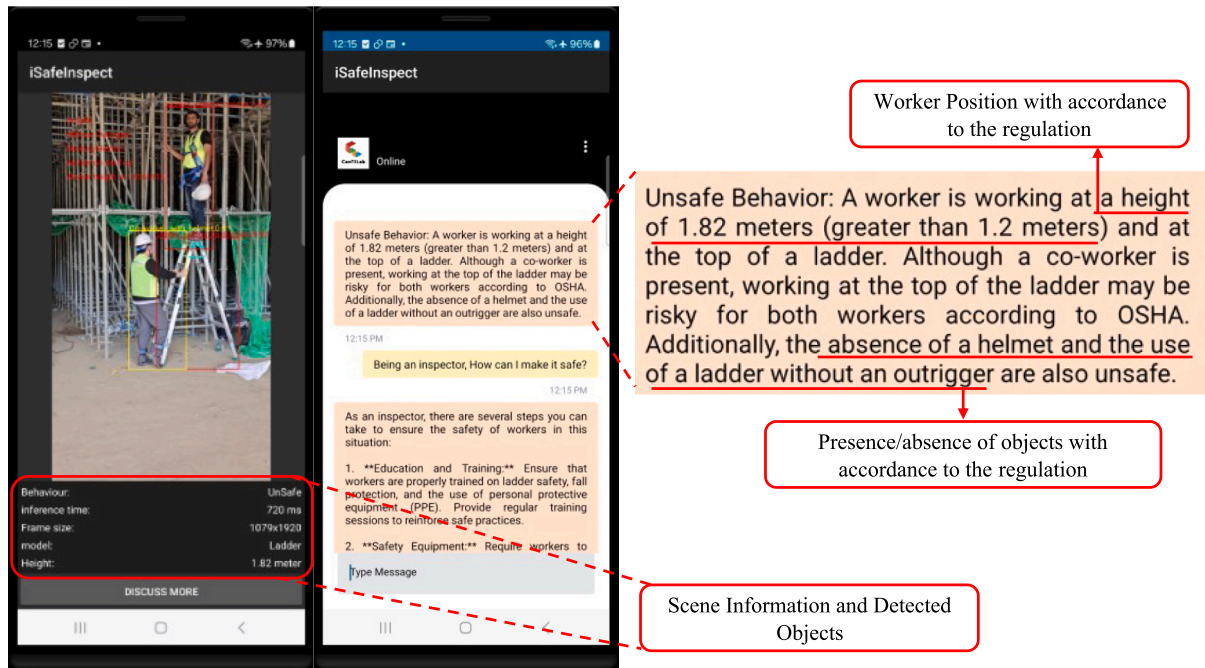
groups: Safety Inspectors, Construction Management Graduates, and Researchers. The radar chart shows high ratings for time-saving, cost-effective, suitable, and comprehensive attributes across all user groups, while lower scores in stability and reliability suggest areas requiring improvement. Primary improvement areas consistently identified across all groups include stability ( $M = 4.73$ .) and reliability ( $M = 4.87$ .), likely attributed to mobile device processing constraints and varying construction site conditions, suggesting focused development needs for



**Table 5**

Classification results against each scenario.

| Scenario/Performance Matrix (%)        | Precision | Recall | F1-Score | Overall Accuracy | Overall Precision | Overall Recall |
|----------------------------------------|-----------|--------|----------|------------------|-------------------|----------------|
| <b>Working on Ladder</b>               |           |        |          | 97.0             | 94.2              | 93.3           |
| Un-Safe                                | 98        | 98     | 98       |                  |                   |                |
| Safe                                   | 94        | 93     | 94       |                  |                   |                |
| <b>Working with Mobile Scaffold</b>    |           |        |          | 94.6             | 98.7              | 87.0           |
| Un-Safe                                | 92        | 99     | 96       |                  |                   |                |
| Safe                                   | 99        | 87     | 93       |                  |                   |                |
| <b>Fire Work (Welding and Cutting)</b> |           |        |          | 99               | 98                | 91             |
| Un-Safe                                | 98        | 100    | 99       |                  |                   |                |
| Safe                                   | 100       | 95     | 97       |                  |                   |                |

**Fig. 9.** Caption detail of unsafe use of ladder case.

dynamic environmental performance. Statistical analysis revealed different confidence intervals ( $95\% \text{ CI} \pm 0.45$  to  $\pm 0.77$ ) indicating acceptable precision, with significant group differences in cost-effectiveness assessment ( $F = 4.161$ ,  $p = .042$ ,  $\eta^2 = 0.410$ ) and system intuitiveness ( $F = 8.216$ ,  $p = .006$ ,  $\eta^2 = 0.578$ ). Researchers demonstrated highest satisfaction with cost-effectiveness and intuitiveness, indicating that technology familiarity influences user acceptance patterns. This clear performance hierarchy indicates the system excels in user interface design and operational efficiency while requiring focused development in dynamic environmental performance—providing critical insights for prioritizing future system enhancements and validating this hybrid CV-LLM approach's current capabilities and limitations.

#### 4. Discussion

The results of this paper illustrate the development, integration, and evaluation of a system designed to enhance safety monitoring in construction environments by leveraging computer vision (CV) and large language models (LLMs). The primary objective is to provide support to inspectors for making decision by the automatic detection and classification of safety hazards while offering real-time, descriptive feedback. Through the validation of the generated captions as well as assessment of user experience, a thorough understanding of the system's performance across different metrics was obtained.

The consistently high mAP scores in detection module performance, across ladder use (97.9 %), mobile scaffolding (97.5 %), and welding

work (98.9 %) validate YOLOv12's attention-centric architecture for complex construction environments. The superior performance on static safety objects versus dynamic human–equipment interactions highlight the challenge of behavioral detection and underscores the system's prioritized reduction of false negatives. The integration of rule-based post-processing with deep learning outputs bridges object detection and regulatory compliance, addressing a key limitation of general-purpose vision models. Minor performance variations between scenarios suggest that scenario-specific fine-tuning and adaptive thresholding are needed for robust field deployment.

Regarding caption generation module evaluation, the analysis reveals that the BLEU scores are generally lower compared to the ROUGE and METEOR scores, primarily due to BLEU's reliance on exact n-gram matches. This can penalize the generated descriptions even when they convey the same meaning using different wording. For instance, in scenario 2, where the generated description identifies "Unsafe conditions detected - A ladder without outriggers poses a safety risk," the BLEU score is low (0.0282) may be due to the differences in the phrases from the ground truths. However, the higher BERT score (0.5489) demonstrates that despite lexical variations, the semantic content closely matches the human evaluations. The relatively high BERT scores in certain cases (e.g., a scenario with a score of 0.5302) suggest that the system is particularly adept at identifying critical safety issues, such as missing helmets or ladders without outriggers, details that are vital for ensuring safety compliance in real-world environments. This capability underscores the system's potential utility in dynamic, safety-critical

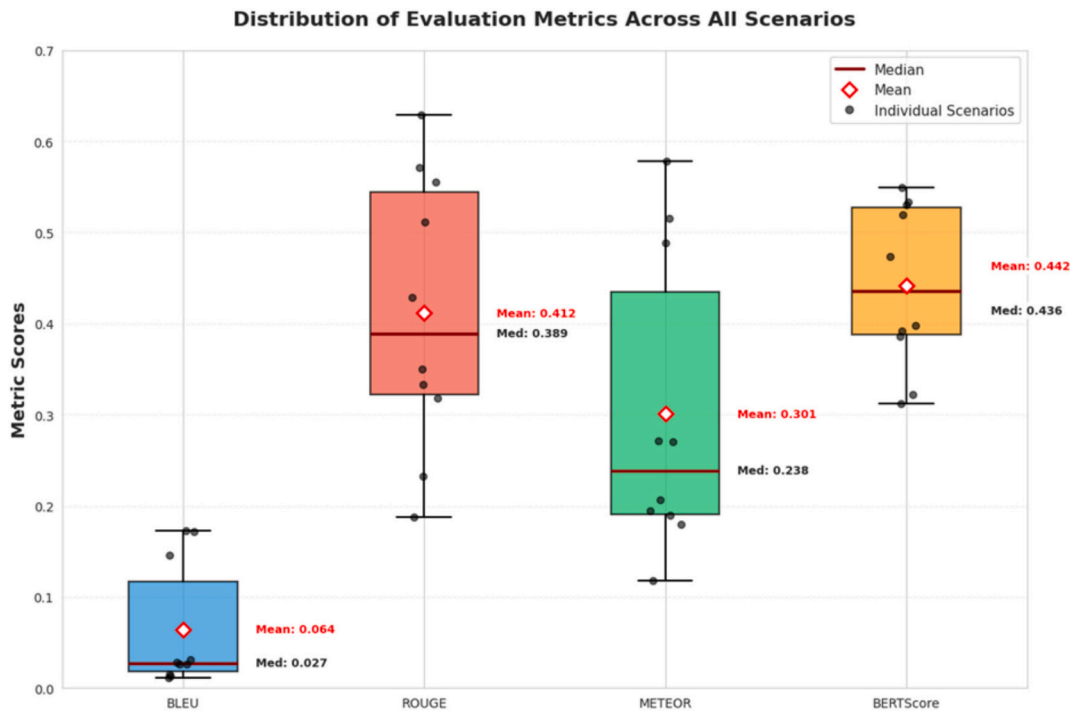


Fig. 10. Distribution of scores for different evaluations across diverse scenarios.

Table 6

Results of user satisfaction against different criteria.

| Criterion                            | Bipolar Adjectives             | Mean | SD*  | CI**         |
|--------------------------------------|--------------------------------|------|------|--------------|
| Reduction in Time and Cost           | Time inefficient - Timesaving  | 6.33 | 0.47 | [5.79, 6.87] |
|                                      | Costly - Cost effective        | 5.87 | 1.05 | [5.18, 6.56] |
| Suitability for Inspection           | Not suitable - Highly suitable | 5.33 | 1.15 | [4.56, 6.11] |
|                                      | Limited - Comprehensive        | 5.67 | 0.89 | [5.09, 6.25] |
| Ease of Use                          | Difficult to use - Easy to use | 6.27 | 0.8  | [5.82, 6.71] |
|                                      | Confusing - Intuitive          | 5.67 | 0.83 | [5.17, 6.16] |
| Operability under Dynamic Conditions | Unstable - Stable              | 4.73 | 0.79 | [3.97, 5.50] |
|                                      | Unreliable - Reliable          | 4.87 | 0.89 | [4.15, 5.59] |
| *SD = Standard Deviation             | CI** = confidence interval     |      |      |              |

contexts where precise identification of hazards is crucial for informed decision-making and timely corrective actions. The ROUGE and METEOR scores across various inputs suggest that the system's generated descriptions generally align with the experts' evaluations, particularly in terms of capturing key safety details. For instance, in scenario 3, the generated description achieves a ROUGE score of 0.5556 and a METEOR score of 0.5156, indicating strong overlap with the ground truths provided by both safety inspectors and researchers. The captioning module was validated using ten visual inputs representing distinct construction tasks. Future work will expand the dataset to include a broader range of scenarios, enhancing the robustness and generalizability of the system while providing more comprehensive insights into captioning performance.

Concerning system effectiveness and user experience evaluation, the results demonstrate a clear performance hierarchy that validates the hybrid CV-LLM approach while identifying specific improvement areas.

The system achieved excellent performance in operational efficiency dimensions (time-saving and ease of use) across all user groups, while demonstrating good performance in functional aspects (cost-effectiveness and comprehensiveness). Researchers demonstrated highest satisfaction with cost-effectiveness and intuitiveness, indicating that technology familiarity significantly influences user acceptance patterns. However, stability and reliability consistently received the lowest ratings across all user groups, which may be attributed to mobile device processing limitations under varying construction site conditions, real-time performance constraints during complex geometric calculations, and dynamic environmental challenges inherent to construction sites. Future improvements may focus on optimizing model inference speed through quantization techniques, implementing robust error handling for network interruptions, enhancing device stability through improved mounting solutions, and developing offline processing capabilities to ensure consistent performance and achieve full deployment readiness for construction safety inspection applications.

This study has several limitations that need to be acknowledged. One primary limitation is the system's reliance on pre-defined safety rules, which may not cover the full spectrum of safety hazards encountered in the ever-changing construction environment. The current system applies task-specific rules (e.g., ladder use) rather than the regulatory corpus. While this approach demonstrates feasibility, it is limited in scope. Future work will focus on integrating comprehensive regulatory databases (e.g., OSHA 1926, KOSHA) to enable broader applicability without manual specification. While the current model is efficient in identifying common risks, construction sites are dynamic and often present unforeseen hazards. Future versions could incorporate adaptive learning mechanisms, enabling the system to refine its hazard detection over time and adapt to evolving safety standards.

The system's performance degrades under partial occlusion, poor illumination, complex backgrounds, and suboptimal camera viewing angles, leading to classification and tracking errors in dynamic multi-worker scenes. Mobile platform constraints also introduced processing delays and network-induced instabilities that impaired real-time monitoring. Moreover, due to the use of GPT API, language modules always need internet connection and sometime produced inconsistent safety

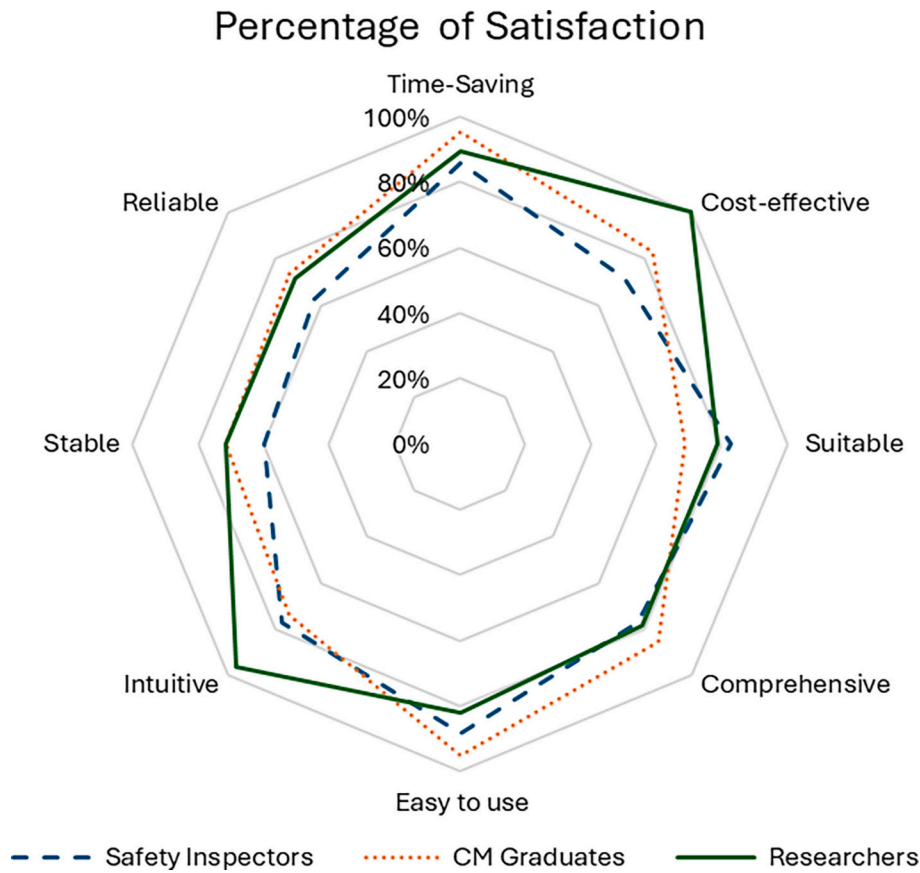


Fig. 11. Satisfaction level of users.

assessments due to ambiguous visual context. Additionally, the height estimation method provides estimated rather than exact physical measurements, with camera perspective, occlusion, and equipment variations introducing deviations of  $\pm 5\text{--}10$  cm error margin, consistent with limitations recognized in prior work by Anjum et al. [28]. Future work will prioritize robust augmentation techniques, optimized inference pipelines, improved multimodal fusion, and enhanced geometric calibration methods to address these measurement uncertainties and enhance overall system reliability and consistency.

Another limitation is the lack of large, diverse datasets specifically curated for construction safety applications. The system's current dataset, while sufficient for preliminary testing, may not encompass the full range of scenarios encountered in global construction projects. The diversity in construction practices across regions suggests the need for a more comprehensive and contextually relevant dataset, to ensure that the proposed system performs consistently across varied construction environments. The computational demands associated with real-time CV and LLM integration are also challenging in some cases. Although the system is designed to operate efficiently on mobile platforms, its scalability and performance across devices with varying computational power remain a consideration. Extensive testing across different hardware configurations will be necessary to ensure that the system can be reliably deployed in diverse on-site conditions. A limitation of the current framework is also its design of static image capture rather than continuous video streams. As a result, rapid transitions from safe to unsafe conditions may not be detected in real time. Future work will focus on extending the framework to handle real-time video analysis and continuous monitoring, enabling faster detection and response in dynamic construction environments. The system remains sensitive to low-quality inputs, underscoring a limitation in data robustness that may affect detection accuracy under unfavorable conditions. Additionally, for the system assessment, the relatively small sample size limits

statistical inference, future studies will expand the participant pool and incorporate additional statistical measures like confidence intervals, effect sizes, and subgroup analyses to increase the credibility of the results. NASA-TLX will be used to assess cognitive workload in future evaluations to further strengthen the rigor of the user experience assessment.

## 5. Conclusion

This paper presented the development and evaluation of an innovative intelligent assistant built on a hybrid reasoning framework that integrates computer vision (CV) and large language models (LLMs) to enhance construction safety management. The system's core contribution lies in its two-stage architecture, which is specifically designed to address the well-documented failure of general-purpose Vision-Language Models to perform the precise, quantitative geometric reasoning required for regulatory compliance. Our framework strategically delegates tasks: a CV module with custom algorithms first performs reliable, rule-based analysis of on-site conditions, and the structured, factual output of this stage is then processed by an LLM for semantic interpretation and interactive, context-aware dialogue.

The system addresses the limitations of traditional safety inspections, which rely heavily on predefined regulations and the subjective judgment of inspectors, by presenting a comprehensive framework to develop a tool that automates hazard detection and facilitates real-time safety discussions. While the current framework requires inspectors to capture images and interact with the chatbot, it reduces subjectivity by providing standardized, regulation-based interpretations. The evaluation of proposed system demonstrated effectiveness in identifying and classifying safety hazards across various scenarios, including ladder use, mobile scaffolding, and fire-related work. The integration of deep learning and LLMs enables the generation of detailed scene descriptions

and contextual safety insights, bridging the gap between visual data capture and actionable safety measures. User feedback and evaluation metrics indicate that the system can enhance efficiency and support informed decision-making in safety inspections. Moreover, future enhancements may include automated generation of draft safety reports during safety walks, which inspectors could review and submit to a centralized repository for broader project team and stakeholder use. This extension would further streamline inspections, improve consistency, and preserve human oversight. The deployment of the system on mobile platforms presents significant potential for the adoption of industry, providing a user-friendly interface that assists safety inspectors. This paper highlights the importance of integrating CV and LLM technologies into construction safety, paving the way for advanced multi-modal information fusion and improved safety outcomes.

Future studies should focus on advancing these systems by enhancing their scalability and customizing them for application in diverse construction contexts. Enhancing the system's ability to analyze complex and dynamic scenarios, such as heavy equipment operation and multi-worker interactions, would broaden its application scope. Integrating real-time risk prediction capabilities using predictive analytics and temporal data from wearable sensors or IoT devices could enable proactive hazard prevention. Furthermore, expanding the multilingual capabilities of the LLM component would facilitate adoption in global construction markets. Collaborative validation with industry stakeholders will be essential to refine the system's usability and ensure alignment with evolving regulatory standards.

Additionally, the incorporation of blockchain technology could be explored to develop an incentive and fine system for promoting safety compliance. Blockchain's transparent and secure framework could reward workers and teams for adhering to safety protocols, reporting hazards, and achieving safety milestones. At the same time, the system could enforce accountability by issuing fines or penalties for violations of safety regulations, such as failing to report hazards or engaging in unsafe practices. This dual approach could foster a culture of proactive safety practices while ensuring strict adherence to safety standards

across organizations. By addressing these areas, the proposed paper can establish itself as a cornerstone for smart safety management not only in construction but also for global industries.

#### CRediT authorship contribution statement

**Rahat Hussain:** Writing – review & editing, Writing – original draft, Validation, Data curation, Conceptualization. **Doyeop Lee:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualization. **Muhammad Sibtain Abbas:** Visualization, Validation, Software, Resources, Data curation. **Syed Farhan Alam Zaidi:** Writing – review & editing, Visualization, Software, Conceptualization. **Akeem Pedro:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Chansik Park:** Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Appendix 1



**Algorithm 2** Ladder Detection and Worker Safety Analysis**Require:** Input Image  $I$ , Detection Results  $R$ **Ensure:** Final Safety Status  $S$ 


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```

1:  $S \leftarrow Safe$ ,  $worker \leftarrow []$ ,  $ladder \leftarrow []$ ,  $reason \leftarrow []$ 
2: for each result  $r \in R$  do
3:   for each  $(x_0, y_0, x_1, y_1, confi, class)$  in  $r$  do
4:     if  $confi > 0.35$  then
5:        $box \leftarrow [x_0, y_0, x_1 - x_0, y_1 - y_0]$ 
6:       if  $class = 0$  then
7:         Append  $box$  to  $ladder$ 
8:       else if  $class = 1$  then
9:          $S \leftarrow Unsafe$ , Append "Ladder without Outriggers" to  $reason$ 
10:        Append  $box$  to  $ladder$ 
11:       else if  $class = 2$  or  $class = 3$  then
12:         Append  $(box, class)$  to  $worker$ 
13:         if  $class = 3$  then
14:            $S \leftarrow Unsafe$ , Append "Worker Without Helmet" to  $reason$ 
15:         end if
16:       end if
17:     end if
18:   end for
19: end for
20:  $highest\_height \leftarrow 0$ ,  $co\_worker \leftarrow 0$ ,  $worker\_height \leftarrow 0$ 
21: Define  $worker\_with\_height \leftarrow []$ ,  $worker\_without\_height \leftarrow []$ 
22: for each  $worker\_box, class$  in  $worker$  do
23:   for each  $ladder\_box$  in  $ladder$  do
24:     if Worker overlaps ladder horizontally then
25:       Compute  $worker\_height$  relative to ladder:

$$worker\_height = \left( \frac{ladder\_box[3] - worker\_box[3]}{ladder\_box[3] - ladder\_box[1]} \right) \times 2$$

26:       if  $worker\_height > highest\_height$  then
27:          $highest\_height \leftarrow worker\_height$ , Insert  $worker$  into  $worker\_with\_height$ 
28:       end if
29:     else
30:       Append  $worker$  to  $worker\_without\_height$ 
31:     end if
32:   end for
33: end for
34: Label workers in  $worker\_without\_height$  based on class, draw bounding boxes
35: for each  $worker$  in  $worker\_with\_height$  do
36:   if First worker then
37:     Label main worker
38:   else
39:     Label as co-worker, Increment  $co\_worker$ 
40:   end if
41: end for
42: if  $highest\_height \geq 1.2$  then
43:   Append "Unsafe height:  $worker\_height$  m" to  $reason$ 
44: end if
45: if  $highest\_height \geq 1.2$  and  $co\_worker = 0$  then
46:    $S \leftarrow Unsafe$ 
47: end if
48: Annotate  $I$  with  $finalStatus$ , reasons if  $S \neq Safe$ 
49: return  $S$ ,  $I$ ,  $reasons$ 

```

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**Algorithm 3** Cutting and Welding Detection**Require:** Input Image  $I$ , Detection Results  $R$ **Ensure:** Final Safety Status  $S$ 


---

```

1: Initialize  $person \leftarrow []$ ,  $hat \leftarrow []$ ,  $final\_status \leftarrow Safe$ ,  $reasons \leftarrow []$ 
2: Set  $saw \leftarrow False$ ,  $fire\_extinguisher \leftarrow False$ ,  $fire\_prevention\_net \leftarrow False$ 
3: for each result  $r \in R$  do
4:   for each bounding box  $(x_0, y_0, x_1, y_1, confi, class)$  in  $r$  do
5:     if  $confi > 0.6$  then
6:       Compute  $box \leftarrow [x_0, y_0, x_1 - x_0, y_1 - y_0]$ 
7:       if  $class = Saw$  then
8:         Set  $saw \leftarrow True$ 
9:       else if  $class = Fire\_Extinguisher$  then
10:        Set  $fire\_extinguisher \leftarrow True$ 
11:       else if  $class = Fire\_Prevention\_Net$  then
12:        Set  $fire\_prevention\_net \leftarrow True$ 
13:       else if  $class = Hard\_Hat$  then
14:        Append  $box$  to  $hat$ 
15:       else if  $class = Person$  then
16:        Append  $box$  to  $person$ 
17:       end if
18:     end if
19:   end for
20: end for
21: for each  $perBox$  in  $person$  do
22:    $hatDetected \leftarrow$  check if hard hat covers the person
23:   if not  $hatDetected$  then
24:      $final\_status \leftarrow Unsafe$  Append "Missing Hardhat" to  $reasons$ 
25:   end if
26: end for
27: if  $fire\_extinguisher$  then
28:   Annotate "Fire Extinguisher: Detected"
29: else
30:    $final\_status \leftarrow Unsafe$ 
31:   Annotate "Fire Extinguisher: Missing"
32:   Append "Fire Extinguisher: Missing" to  $reasons$ 
33: end if
34: if  $saw$  then
35:   if  $fire\_prevention\_net$  then
36:     Annotate "Cutting: Fire Prevention Net detected"
37:   else
38:      $final\_status \leftarrow Unsafe$ 
39:     Annotate "Cutting: Fire Prevention Net Missing"
40:     Append "Cutting: Fire Prevention Net Missing" to  $reasons$ 
41:   end if
42: end if
43: Annotate image with  $final\_status$ 
44: return  $S, I, reasons$ 

```

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**Algorithm 4** Mobile Scaffolding Detection**Require:** Input Image  $I$ , Detection Results  $R$ **Ensure:** Final Safety Status  $S$ 

```

1: Initialize lists:  $worker\_with\_helmet \leftarrow []$ ,  $worker\_without\_helmet \leftarrow []$ ,  $mobile\_scaffold\_outrigger \leftarrow []$ 
2: Set  $final\_status \leftarrow Safe$ ,  $final\_message \leftarrow []$ 
3: for each result  $r \in R$  do
4:   for each bounding box  $(x_0, y_0, x_1, y_1, confi, class)$  in  $r$  do
5:     if  $confi > 0.6$  then
6:       Compute  $box \leftarrow [x_0, y_0, x_1 - x_0, y_1 - y_0]$ 
7:       if  $class = 0$  then
8:         Append "Missing Guardrail" to  $final\_message$ , Set  $final\_status \leftarrow UnSafe$ 
9:       else if  $class = 1$  then
10:        Append "No Outrigger" to  $final\_message$ , Set  $final\_status \leftarrow UnSafe$ 
11:        Append  $box$  to  $mobile\_scaffold\_outrigger$ 
12:       else if  $class = 2$  then
13:        Append  $box$  to  $mobile\_scaffold\_outrigger$ 
14:       else if  $class = 3$  then
15:        Append  $box$  to  $worker\_with\_helmet$ 
16:       else if  $class = 4$  then
17:        Append "Missing Helmet" to  $final\_message$ , Set  $final\_status \leftarrow UnSafe$ 
18:       end if
19:     end if
20:   end for
21: end for
22:  $total\_no\_person\_on\_scaffolding \leftarrow 0$ 
23: for each scaffold in  $mobile\_scaffold\_outrigger$  do
24:   for each person in  $worker\_with\_helmet$  do
25:     if Person is inside scaffold bounds then
26:       Increment  $total\_no\_person\_on\_scaffolding$ 
27:     end if
28:   end for
29: end for
30: if  $total\_no\_person\_on\_scaffolding > 2$  then
31:   Append " $total\_no\_person\_on\_scaffolding$  Worker" to  $final\_message$ 
32:   Set  $final\_status \leftarrow UnSafe$ 
33: end if
34: Annotate  $I$  with  $final\_status$  and  $final\_message$ 
35: return  $S, I, final\_message$ 

```

**Data availability**

Data will be made available on request.

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